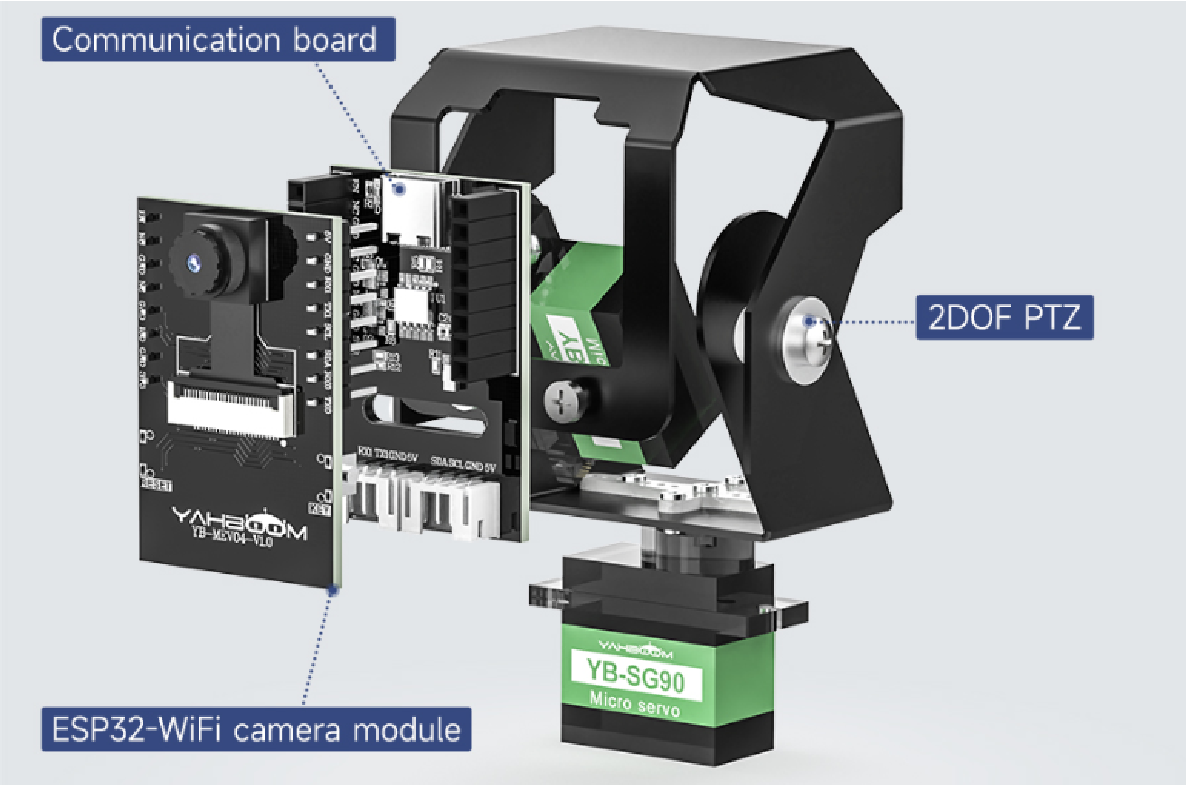
ESP32 Vision Recognition Module Introduction

The ESP32-WiFi image transmission module is a compact, cost-effective AI vision solution designed for rapid integration and development. The module adopts a dual-board architecture consisting of a core board and a communication board, with an optional 2-DOF pan-tilt vision head. The core board features an onboard ESP32-S3 chip, an integrated camera, pin headers, and a custom button, supporting AI features such as WiFi image streaming, face recognition, and color recognition. Factory firmware is pre-installed for immediate use.

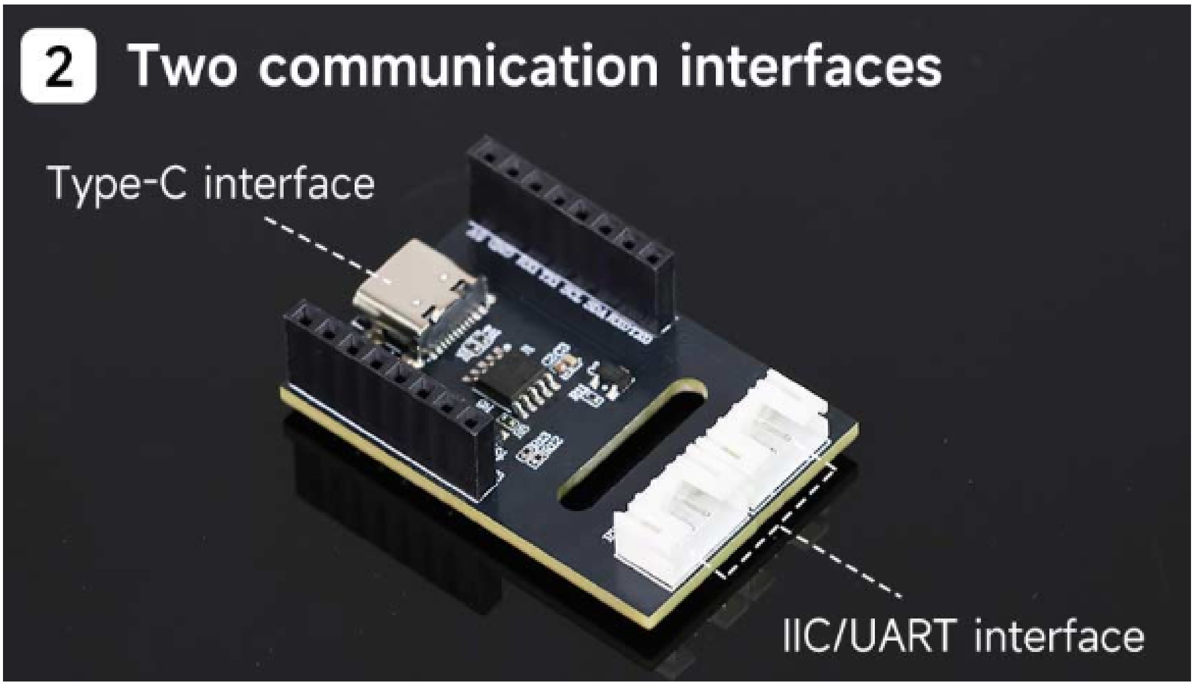
On the robot car, the WiFi image transmission module streams high-definition images back to the PC virtual machine in real time with low latency and high stability, providing a high-quality visual input stream for large models and AI algorithms. It easily unlocks cutting-edge interactive features such as face recognition, color tracking, and multimodal large-model visual analysis, broadly expanding the boundaries of the robot's visual perception.

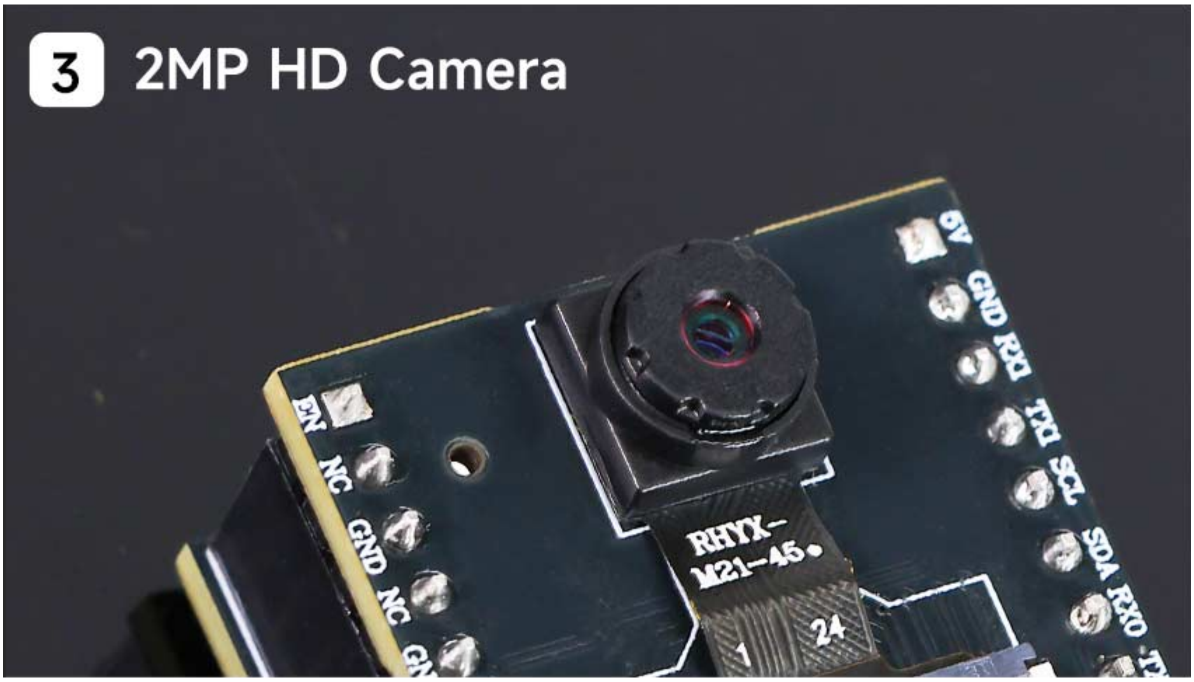


1. Hardware Features

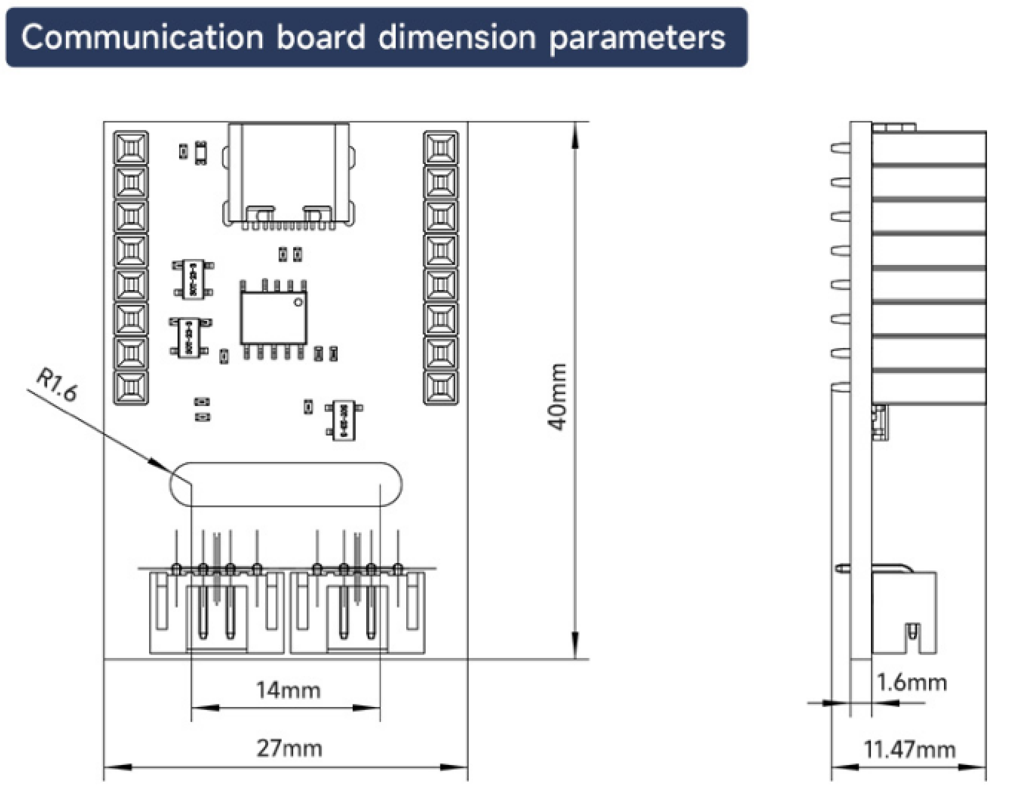
- Uses the ESP-32 high-performance chip: a low-power dual-core 32-bit CPU running at 240MHz, with computing power up to 600MIPS
- Onboard Type-C interface and IIC/UART interface, supporting two communication methods
- Onboard a new high-definition camera supporting 2MP (200W) output. The image is clear, based on CMOS image sensor technology with high sensitivity and low power consumption
- Onboard reset button and a customizable programming button to meet the user's custom programming needs
- The ESP32-S3 WiFi image transmission module Lite version is compact, measuring only 2.7x4cm, suitable for diverse scenarios and enabling more creative AI projects


1
ESP32 High-Performance Chip
ESPRESSIF
ESP32-S3-WROOM-1U
MCN4R2


2
Two communication interfaces
Type-C interface
IIC/UART interface

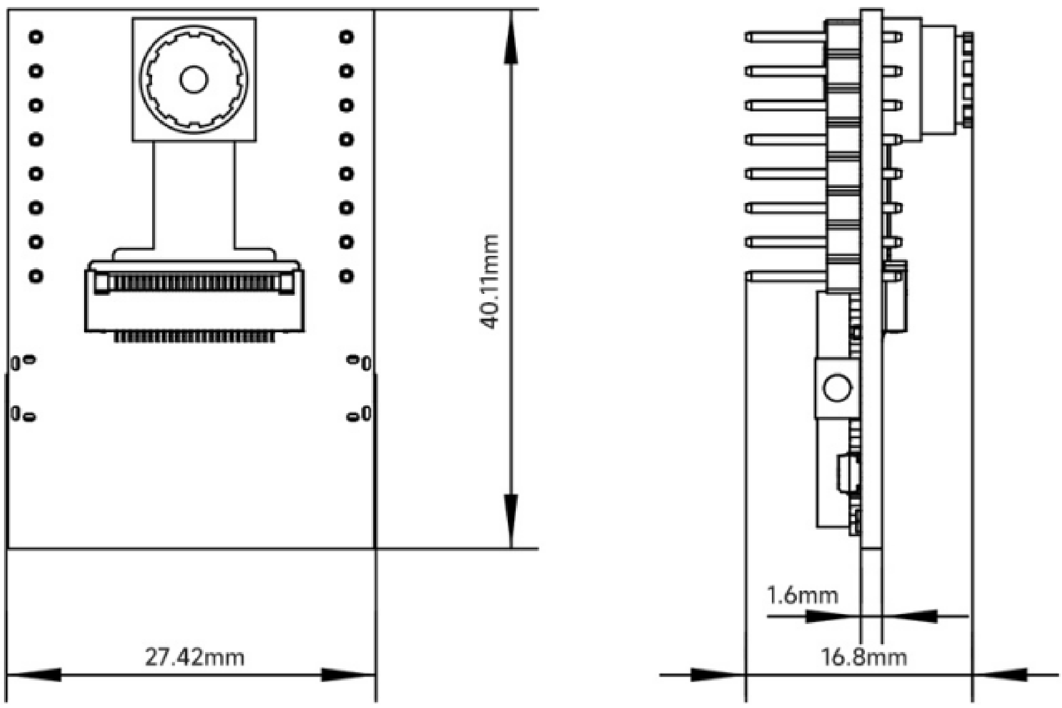


2. Dimensions and Parameters



Serial port interface definition	**5V power positive TX serial transmission GND power negative RX serial reception**	**4-pin connector type**	**PH2.0**
IIC interface definition	5V power positive SDA interface SCL interface GND power negative	Dimensions	40mm*27mm*1.6mm
Female header pitch	2.54mm	Weight	7.8g
Female header length	8.5mm		

ESP32-WiFi camera module lite version dimension parameters



Antenna model	**2.4GHz FPC (flexible board) antenna**	**Button**	**Custom button, RESET button**
Antenna gain	3dB gain	Working mode	Default AP mode + STA mode
Transmission distance	≤20m	Dimensions	40mm*27mm*1.6mm
Supply voltage	5V	Weight	7.8g
Supply current	250mA		